Pneumonia Detection from Chest X-Rays

A Machine Learning Approach

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Significance of the Problem

Pneumonia is a leading infectious cause of mortality worldwide, responsible for over **2.5 million deaths annually**, disproportionately affecting children under five and the elderly. Diagnosis traditionally relies on radiologists manually inspecting chest X-rays — a process that is time-consuming, costly, and prone to error, especially in resource-limited settings. Automating detection via machine learning can accelerate clinical decisions, reduce workload, and extend diagnostic capabilities to underserved regions where specialist access is scarce.

Challenges

- **Class Imbalance:** The dataset is heavily skewed toward pneumonia cases ($\sim$74% of training data), causing models to be biased toward the majority class and inflate accuracy while misclassifying healthy patients.
- **Feature Engineering without Deep Learning:** Restricted to classical ML (no CNNs), meaningful features must be manually crafted from raw pixels (statistics, histograms, texture descriptors), which is non-trivial given image complexity.
- **High Dimensionality:** Flattening X-ray images into vectors yields thousands of features, risking the curse of dimensionality and requiring careful reduction (e.g., PCA).
- **Evaluation:** Standard accuracy is misleading under imbalance; F1-score, AUC-ROC, and minority-class recall must be prioritized.

Type of Problem

This is a **supervised binary classification** problem in **medical image analysis using classical machine learning**. The goal is to classify a chest X-ray as *Normal* or *Pneumonia* using handcrafted features — without convolutional networks or deep learning.

Inputs and Outputs

Input: A grayscale chest X-ray image resized to a fixed resolution (e.g., 64×64), then transformed into a feature vector via pixel intensity statistics (mean, variance, skewness), intensity histograms, and PCA. *Example:* A 1024×768 JPEG is resized, flattened, and compressed to a 100-dimensional PCA vector.

Output: A binary label — **0** = *Normal* / **1** = *Pneumonia*.

Dataset Description

Dataset: Chest X-Ray Images (Pneumonia) **Source:** kaggle.com – Paul Mooney

The dataset contains **5,863 JPEG chest X-ray images** split across three sets:

Split	Normal	Pneumonia	Total
Train	1,341	3,875	5,216
Validation	8	8	16
Test	234	390	624

Imbalance Note: Pneumonia cases constitute $\sim$**74%** of training images, requiring mitigation via SMOTE, undersampling, or class-weighted classifiers. The validation set (only 16 images) should be merged with training data.

Preprocessing Pipeline

1. **Image Standardization:** All images resized to 64×64 pixels (down from original resolutions up to 1500×1500) to balance detail preservation with computational efficiency. Bilinear interpolation used for resizing.
2. **Intensity Normalization:** Pixel values converted to float32 and scaled to $[0, 1]$ range via $x_{norm} = (x - x_{min})/(x_{max} - x_{min})$, ensuring uniform input distribution across all samples.
3. **Feature Extraction:** Four complementary feature sets extracted:
   - *Statistical Features:* Mean, variance, skewness, kurtosis, and percentiles (25th, 50th, 75th) per image
   - *Texture Features:* Gray-level co-occurrence matrix (GLCM) with contrast, correlation, energy, homogeneity
   - *Histogram Features:* 32-bin intensity histogram capturing global intensity distribution
   - *Edge Features:* Canny edge detection statistics (edge density, mean edge intensity)

   Total feature dimension after extraction: **89 features per image**.
4. **Dimensionality Reduction:** Principal Component Analysis (PCA) applied with 95% variance retention, reducing features to $\sim$25-30 components while preserving discriminative information.
5. **Class Imbalance Handling:** Synthetic Minority Over-sampling Technique (SMOTE) applied to training data only, generating synthetic normal samples to achieve 1:1 class ratio. k-neighbors parameter set to 5 for interpolation.
6. **Train-Validation Split:** Original validation set (16 images) merged with training data; new stratified 80-20 split creates validation set of $\sim$1,043 samples preserving class distribution.